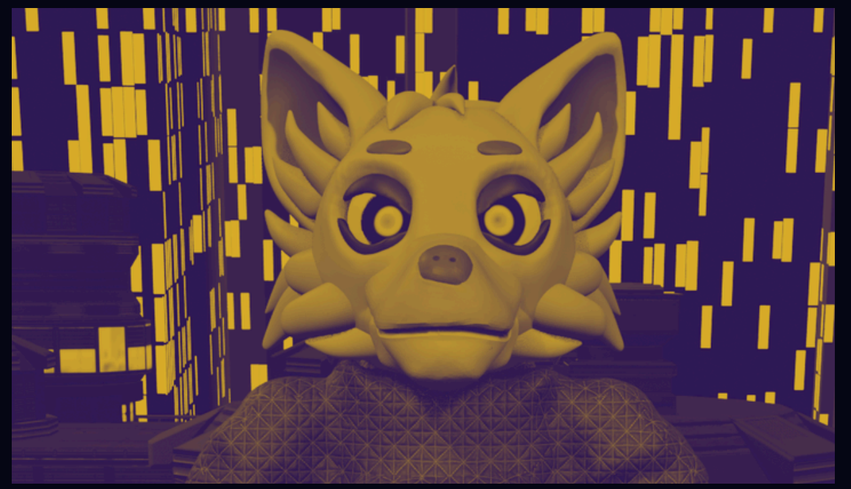
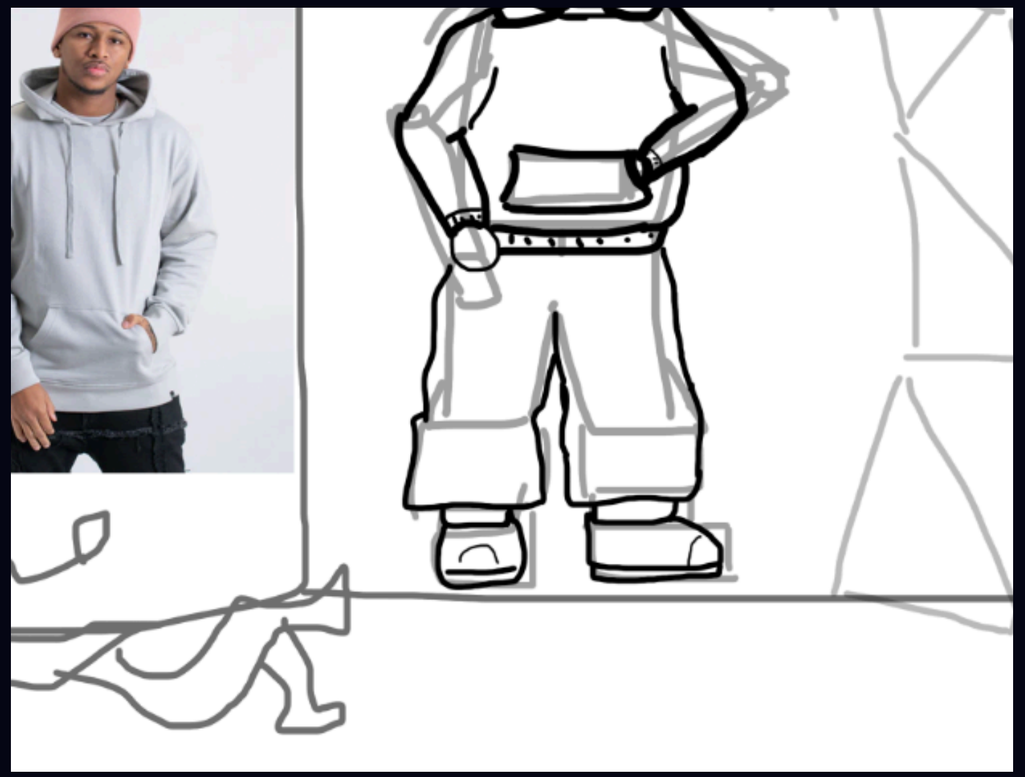
DESIGN AND DEVELOPMENT OF ART TOY

ANIMHOODIE

Your Name: Natnithichai Srikolchan
Student ID: 633040585-1
Advisor: Dr. Sarun Paisarnsrisomsuk
Co-Advisor: Asst. Prof. Manasawee Kaenampornpan





PROJECT OVERVIEW

What is Anim Hoodie?

- Stylized 3D art toy combining streetwear aesthetics with character design
- Focus on simplicity, individuality, and modern artistic expression

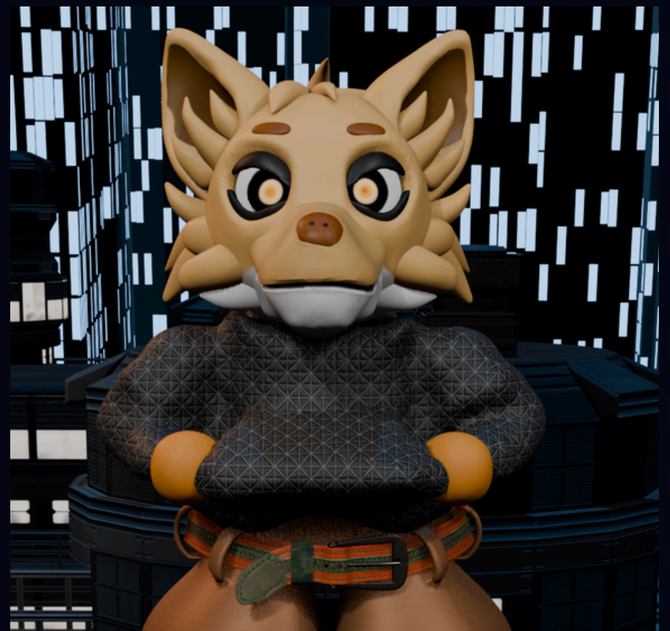
Project Scope:

- Complete digital production pipeline
- From concept sketches to final high-quality renders

PROJECT OBJECTIVES

Core Goals:

- Design unique 3D art toy with strong visual identity
- Master professional 3D modeling and sculpting workflow
- Apply advanced texturing and material techniques
- Execute high-quality rendering and lighting
- Demonstrate complete production pipeline



TOOLS & SOFTWARE

Primary Software:

Blender3D

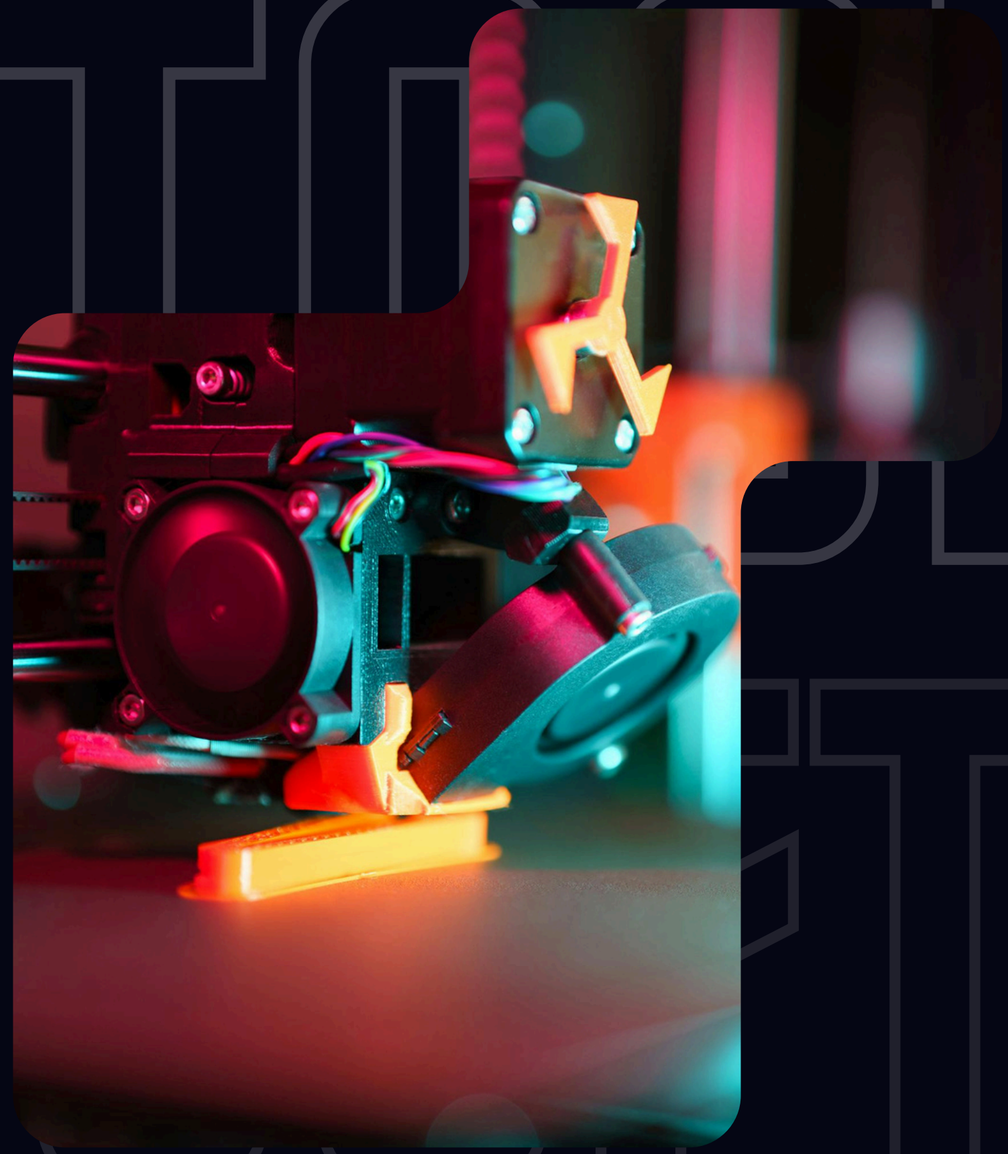
- modeling and base mesh creation
- Digital sculpting and detail work
- Texturing and material setup
- Lighting and rendering

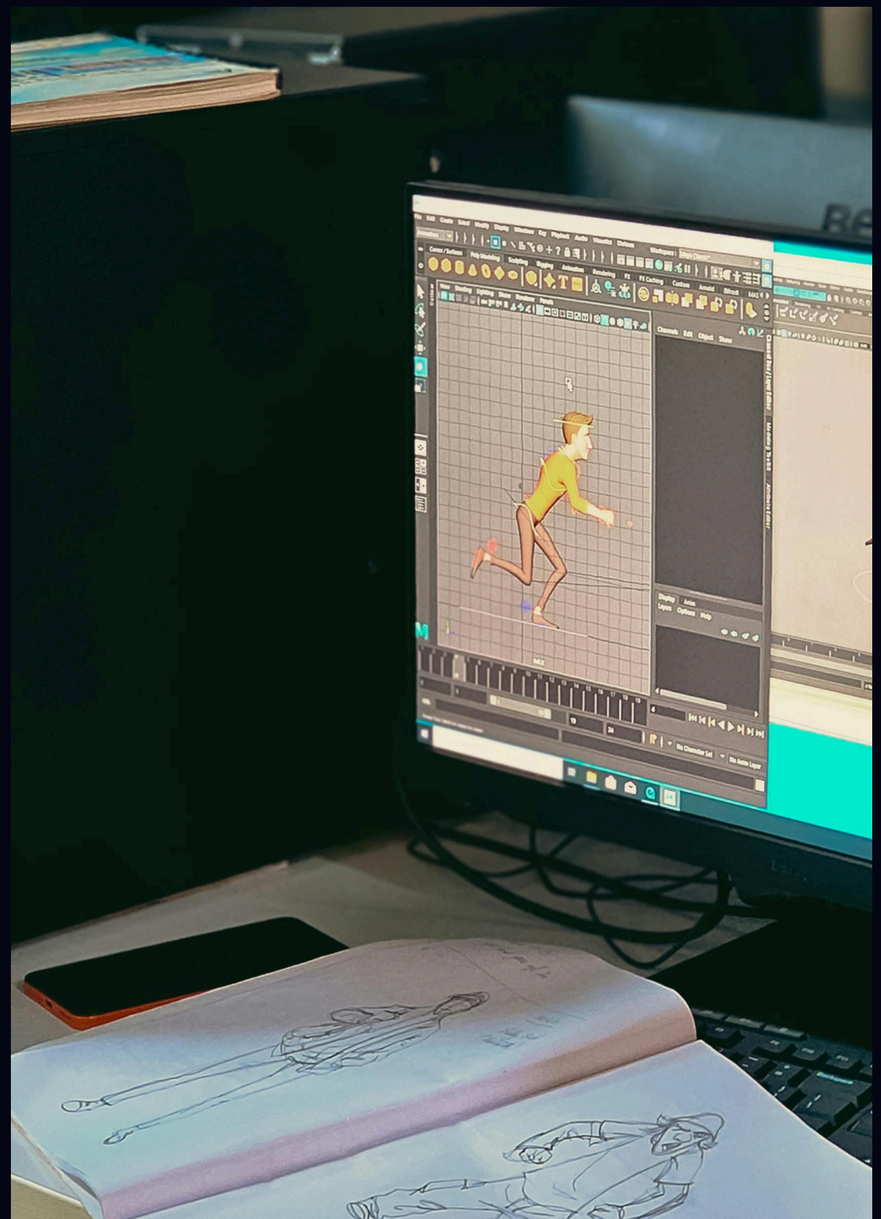
Adobe Photoshop

- Concept sketching and refinement
- Texture map editing
- Post-processing and final composition

Hardware:

- High-performance computer with GPU acceleration
- Drawing tablet for concept work





COMPLETE PRODUCTION PIPELINE

7-Stage Workflow:

1. RESEARCH & IDEATION
↓
2. CONCEPT SKETCHING
↓
3. BASE MESH CREATION
↓
4. DIGITAL SCULPTING
↓
5. RETOPOLOGY & UV UNWRAPPING
↓
6. TEXTURING & MATERIAL SETUP
↓
7. LIGHTING & RENDERING

STAGE 1
RESEARCH & IDEATION

01 **Research Phase:**

- Analyzed successful art toy brands (Pop Mart, Mighty Jaxx, 52TOYS)
- Studied design trends and aesthetics
- Gathered visual references from multiple sources

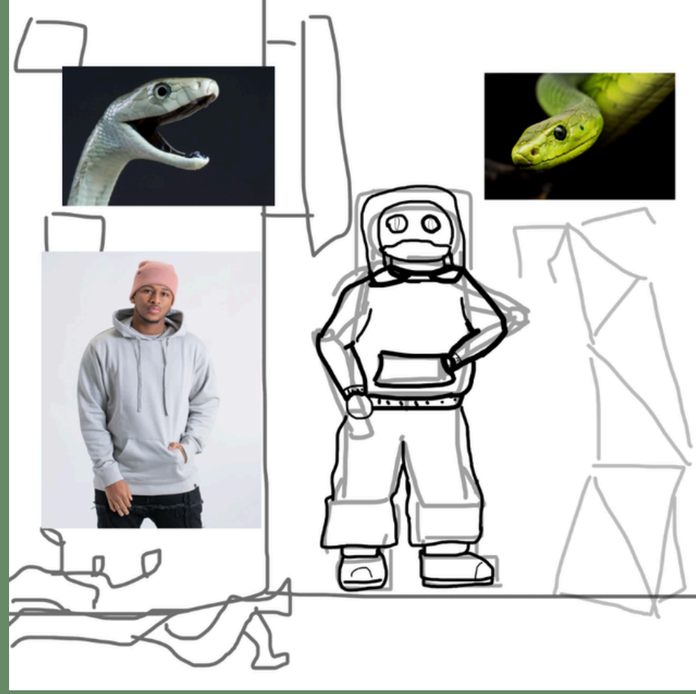
02 **Key Findings:**

- Strong silhouettes create memorability
- Simplicity enhances collectible appeal
- Character personality drives emotional connection

03 **Research Focus:**

- Streetwear fashion aesthetics
- Animal character stylization
- Collectible toy proportions

STAGE 2 INITIAL CONCEPT EXPLORATION



01

Step 1: Visual Exploration

- Compiled moodboards and references
- Explored different animal characters
- Tested various poses and expressions

02

Step 2: Character Direction

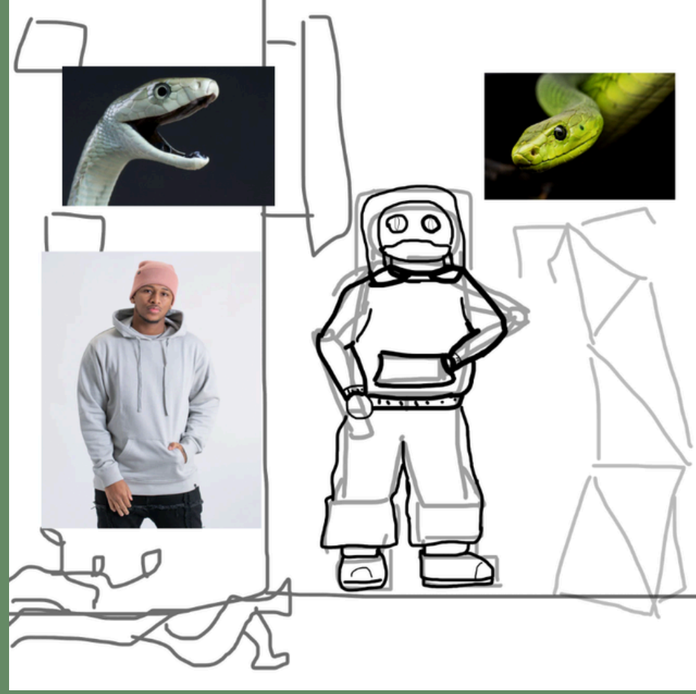
- Strong silhouettes create memorability
- Simplicity enhances collectible appeal
- Character personality drives emotional connection

03

Design Principles:

- Clean, recognizable silhouette
- Emotional expressiveness
- Toy-like proportions

STAGE 2
SKETCH REFINEMENT



01

Multiple Iterations Exploring

- Character posture and stance
- Proportions and body structure
- Facial features and expression
- Accessory placement

02

Software:

Adobe Photoshop with drawing tablet

03

Goal: Achieve balance between:

- Visual appeal
- 3D modeling feasibility
- Production simplicity

STAGE 2
FINAL CONCEPT APPROVAL

01

Technical Preparation:

Created Turnaround Sheet:

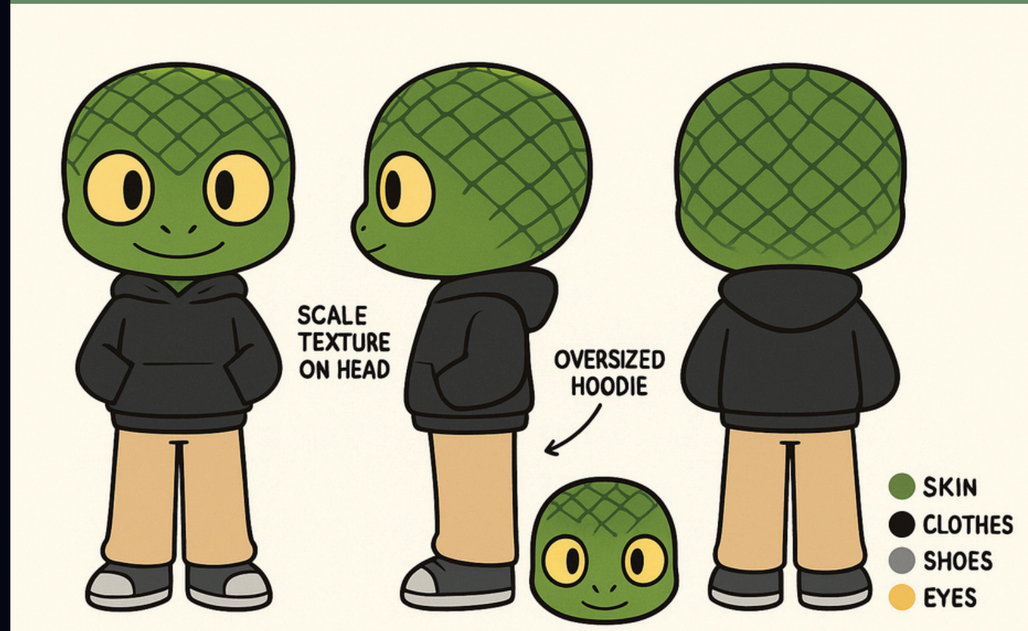
- Front view (primary reference)
- Side view (profile structure)
- Back view (complete understanding)

02

Key Decisions:

- Neutral color palette (focus on form)
- Clear structural lines
- Defined proportions and measurements

Ready for 3D Translation



STAGE 3
BASE MESH CREATION (START)

01

Starting Point:

- Begin with primitive shapes (spheres, cubes, cylinders)
- Block out basic character proportions
- Establish overall silhouette

02

Initial Focus:

- Correct body-to-head ratio
- Proper limb proportions
- Balanced stance and posture

STAGE 3
BASE MESH CREATION (PROCESS)

01

Modeling Workflow:

- Scale and position primitive shapes
- Join and merge meshes
- Apply Mirror Modifier for symmetry
- Add Subdivision Surface for smoothness

02

Critical Considerations:

- Clean edge flow for future sculpting
- Proper topology around joints
- Maintain reference to concept sketches

STAGE 3
BASE MESH REFINEMENT

01

Adjustments Made:

- Refine proportions to match concept
- Adjust volume distribution
- Create smooth transitions between body parts
- Establish primary forms (body, head, hoodie, limbs)

02

Quality Check:

- Silhouette matches concept from all angles
- Proportions feel balanced
- Mesh ready for detailed sculpting

STAGE 4
DIGITAL SCULPTING (INTRODUCTION)

01

Sculpting Objectives:

- Add surface detail and character
- Define clothing folds and wrinkles
- Enhance facial features and expression
- Create realistic surface topology

02

Blender Sculpting Tools Used:

- Clay Buildup - Adding volume
- Crease - Sharp edges and folds
- Smooth - Blending transitions
- Grab - Adjusting large forms
- Inflate - Adding rounded volume

STAGE 4 SCULPTING PROCESS (PART 1)

01

Primary Sculpting Phase:

1. Increase mesh subdivision for detail
2. Define major forms (hoodie shape, body curves)
3. Add clothing folds and wrinkles
4. Sculpt facial features and expression

02

Design Approach:

- Emulate vinyl toy aesthetic
- Balance smoothness with detail
- Consider how light will interact with surfaces

STAGE 4 SCULPTING PROCESS (PART 2)

01

Secondary Sculpting Phase:

- Refine fabric texture
- Add subtle surface variations
- Enhance edge definition
- Polish overall form quality

02

Artistic Decisions:

- Maintain toy-like simplicity
- Avoid over-detailing
- Focus on clean, readable surfaces

Checkpoint:

Save multiple versions to compare progress

STAGE 5
RETOPOLOGY & UV UNWRAPPING

01

Retopology Process:

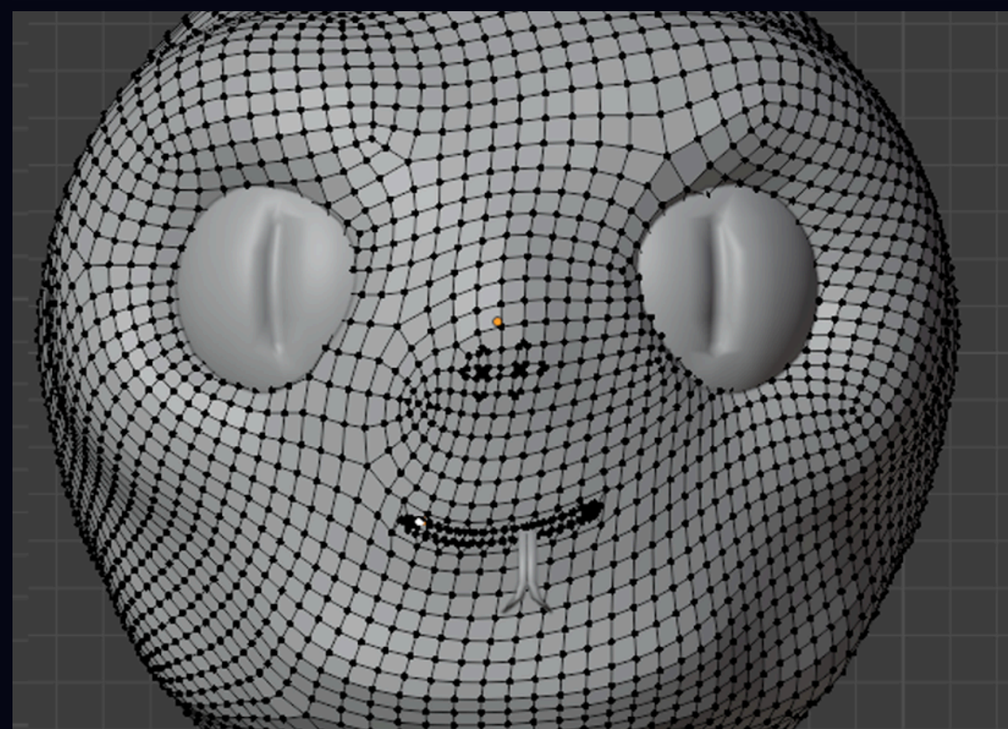
- Create clean, low-poly mesh over sculpt
- Reduce millions of polygons to manageable count
- Maintain visual quality

02

UV Unwrapping:

- Convert 3D surface to 2D texture map
- Minimize stretching and distortion
- Strategic seam placement

Result: Production-ready optimized mesh



STAGE 6
TEXTURING & MATERIALS

01

In Blender:

- Base color painting and application
- Material map creation (Roughness, Normal, Metallic)
- Shader setup and node configuration

02

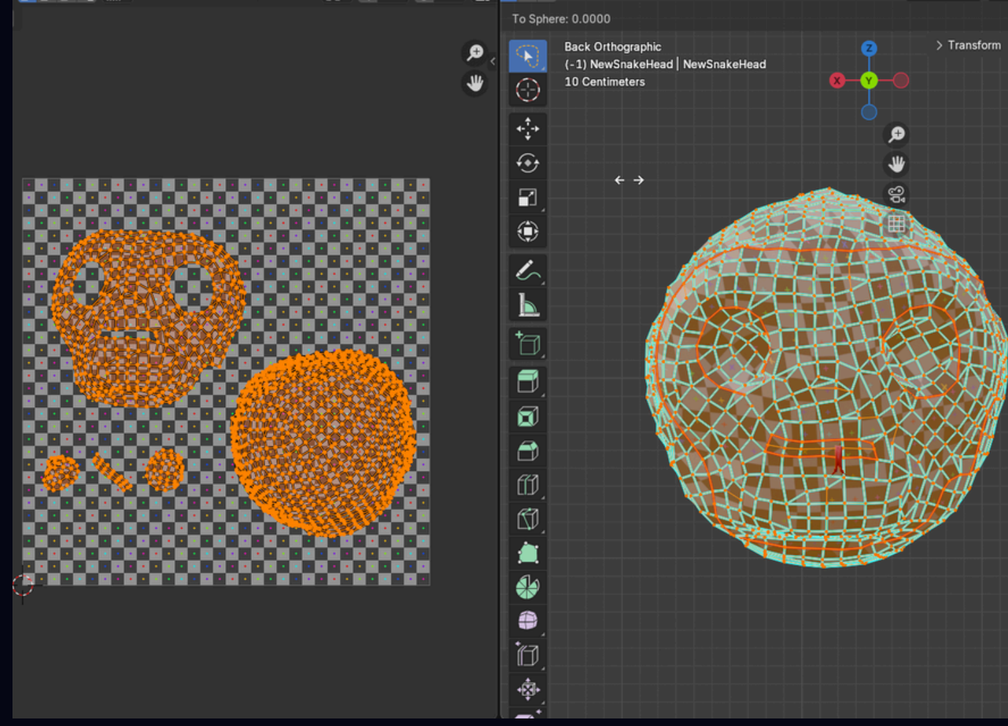
In Photoshop:

- UV map painting and refinement
- Color adjustments and details
- Texture map editing

03

Design Approach:

- Soft neutrals and muted tones
- Vinyl toy aesthetic
- Clean, minimal details



CHARACTER VARIATIONS COMPLETE PIPELINE

Three Design Iterations:

Snake Design (Original)

Fox Design (Alternative)

Capybara Design (Additional)

Each Shows Complete Process:

Base Mesh → Sculpting & Materials → Final Shaded Result

Demonstrates:

- Workflow versatility
- Consistent quality
- Repeatable process

STAGE 7
LIGHTING & RENDERING SETUP

Lighting Configuration:

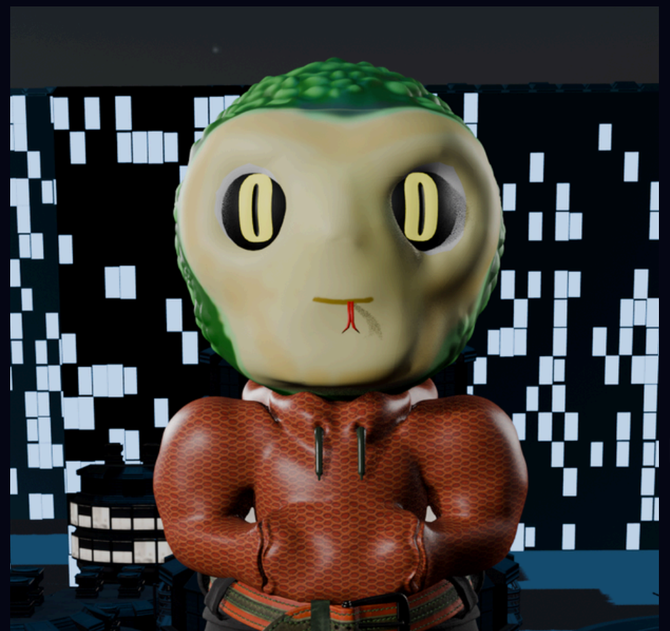
- HDRI environment mapping
- Three-point lighting system
- Studio photography simulation

Render Settings:

- 1080 resolution (1920 × 1080)
- Blender Eevee engine
- 16-100 samples
- GPU acceleration

Camera Work:

- Multiple angles prepared
- Focus on character personality
- Professional composition



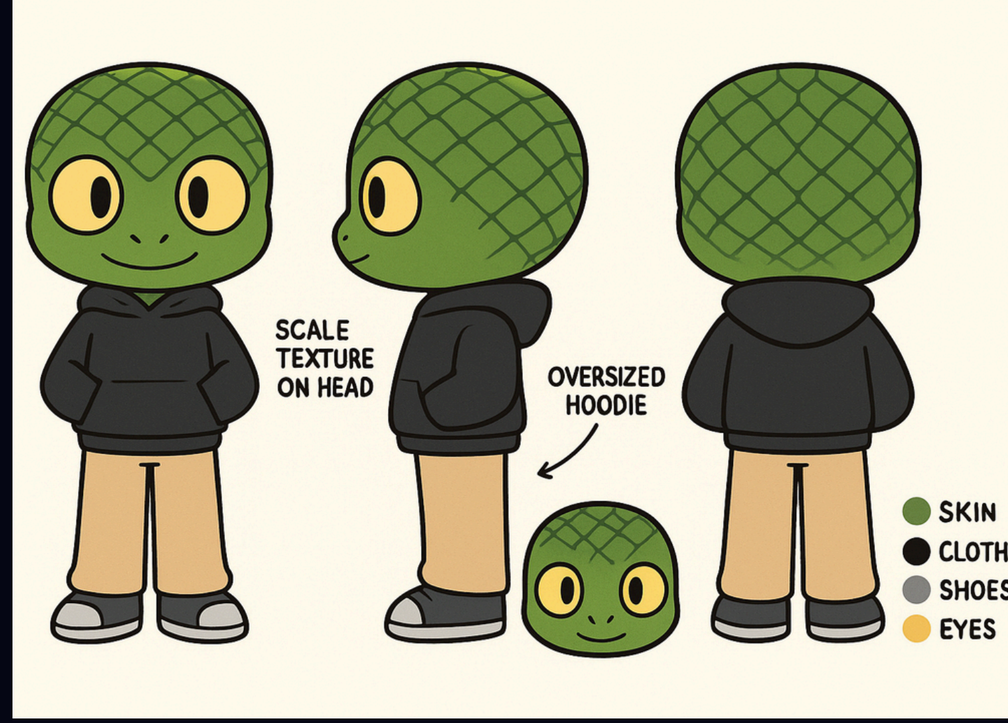
STAGE 7
POST-PROCESSING

01

Photoshop Adjustments:

- Levels and curves
- Sharpness enhancement
- Contrast refinement
- Color grading
- Subtle depth of field

Goal: Polish to portfolio-ready quality



FINAL RESULTS
FRONT VIEW

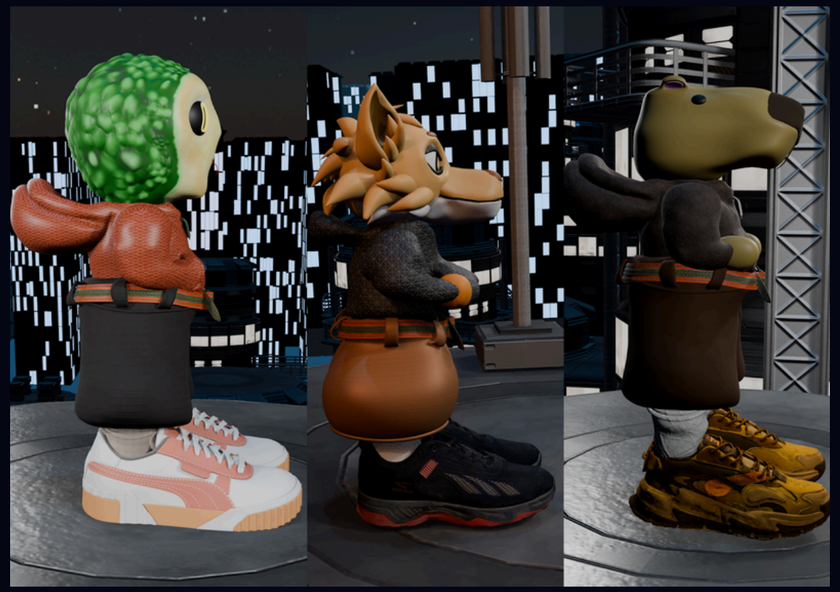
Showcases:

- Professional lighting quality
- Realistic material rendering
- Character personality
- Clean composition

Technical Achievement:
Complete 7-stage pipeline successfully executed

FINAL RESULTS MULTIPLE ANGLES

Back View:
Consistent quality throughout
Complete modeling details



Side Views:
Proper 3D form and depth
Balanced proportions

Close-Up:
Material and texture quality
Surface detail excellence

PROCESS TRANSFORMATION COMPLETE EVOLUTION:

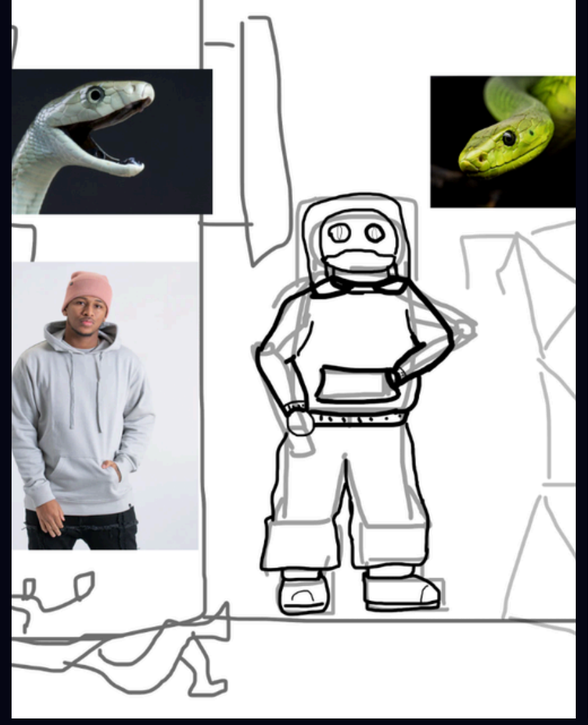
5-Stage Visual Journey:

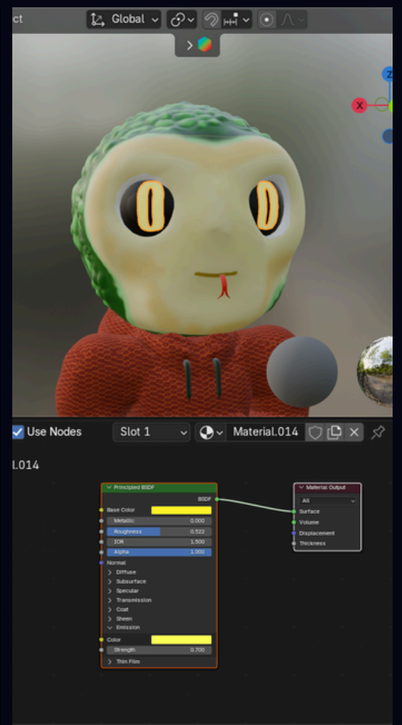
Concept Sketch → Base Mesh → Sculpted Model → Textured → Final Render

Timeline: 8 months of systematic development

Shows: Professional progression from idea to execution

Visuals: Linear progression across slide showing all stages





TECHNICAL WORKFLOW SUMMARY

Production Pipeline Overview:

Stage	Duration	Key Output
Research & Concept	4 weeks	Approved designs
Base Mesh	2 weeks	Clean topology
Sculpting	4 weeks	Detailed model
Retopo & UV	3 weeks	Optimized mesh
Texturing	4 weeks	Materials applied
Rendering	2 weeks	Final images

Total Active Production: ~19 weeks

KEY CHALLENGES & SOLUTIONS

Problem-Solving Throughout

Challenge 1:
Proportion Balance → Solution: Multiple iterations, peer feedback

Challenge 2:
Detail Management → Solution: Reference study, controlled refinement

Challenge 3:
Topology Optimization → Solution: Strategic retopology planning

Challenge 4:
Render Efficiency → Solution: Settings optimization, GPU usage

Learning: Planning prevents problems

PROJECT ACHIEVEMENTS

Successfully Completed:

Design Excellence

- Original character design
- Three complete variations
- Consistent quality

Technical Mastery

- Full production pipeline
- Professional-grade output
- Industry-standard workflow

Portfolio Quality

- High-resolution renders
- Multiple presentation angles
- Complete documentation

THANK YOU

Special Thanks:
Dr. Sarun Paisarnsrisomsuk
Asst. Prof. Manasawee Kaenampornpan
Faculty of Engineering, Khon Kaen University
Peers and reviewers

Project: Design and Development of Art Toy "Anim Hoodie"
Date: July 2025

Questions & Discussion Welcome